

Number Path Subtraction

Answer Key

Kindergarten–Grade 1

Quick Answers

- | | |
|--------------------------|--------------------------|
| 1. 5 | 6. 3 |
| 2. 3 | 7. 6 |
| 3. the difference = 5, — | 8. 5 |
| 4. 7 | 9. the difference = 3, — |
| 5. 2 | 10. how many hops = 4, — |
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What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 3, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

1.OA.C.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 13 tagged checks.

1. Here is the pattern. Start at 7. Hop back 2: you land on 6, then 5. So $7 - 2 = 5$. Now trace it yourself and write the answer.

Solution: Finger on 7, hop back two boxes: 6, then 5. Two hops for the 2 being subtracted. So $7 - 2 = 5$. 5

2. Start at 4. Hop back 1.

Solution: Finger on 4, one hop back lands on 3. So $4 - 1 = 3$. 3

3. Draw your hops on the path: start at 8 and hop back 3.

Solution: Finger on 8, hop back three boxes: 7, 6, 5. So $8 - 3 = 5$. 5

Hops (instructor judges): three arcs drawn from 8 landing on 7, 6, and 5.

4. Use the path to help you. $9 - 2$

Solution: Start at 9, hop back twice: 8, 7. So $9 - 2 = 7$. 7

5. A new way! To find $6 - 4$, start at 4 on the path and count the hops *up* to 6.

Solution: Finger on 4, hop up: 5, 6 — that is 2 hops to reach 6. The number of hops is the difference, so $6 - 4 = 2$. 2

6. Start at 7. Count the hops up to 10. $10 - 7$

Solution: From 7, hop up: 8, 9, 10 — three hops. So $10 - 7 = 3$.

3

7. Pick your way: hop back, or count up. $10 - 4$

Solution: Hopping back 4 from 10 lands on 6. (Counting up from 4 to 10 takes six hops — same answer, more hops!) So $10 - 4 = 6$.

6

8. The hop number is hiding! Start at 8. You land on 3. How many hops back did you make? $8 - \square = 3$

Solution: Count the boxes from 8 down to 3: 7, 6, 5, 4, 3 — that is 5 hops. Check: $8 - 5 = 3$. The hidden hop number is

5

9. Pick your way: hop back 6 times, or count up from 6. Draw your hops on the path. $9 - 6$

Solution: Counting up is quicker here: from 6, hop 7, 8, 9 — only 3 hops instead of six back. So $9 - 6 = 3$.

3

Hops (instructor judges): either six arcs back from 9 landing on 3, or 3 arcs up from 6 to 9; the hop count must match the answer written.

10. Challenge! Mo starts at 10 and hops back. He lands on 6. How many hops did he make? Tell how the path shows it. $10 - \square = 6$

Solution: Count the boxes from 10 down to 6: 9, 8, 7, 6 — that is 4 hops. Check: $10 - 4 = 6$. The missing hop number is

4

Tell how (instructor judges): a good answer says the hops between 10 and 6 can be counted on the path — 4 hops either direction — so the missing number is 4.